

Minimum Ratio Test

- Canonical form

- Def: Consider a linear system $Ax=b$, where A is $m \times n$ and $\text{rank}(A)=m < n$. The linear system is said to be in canonical form if A contains I_m as a submatrix, where I_m is the $m \times m$ identity matrix.

e.g.,

$$\begin{array}{rcl} x_1 - 2x_2 + x_3 & & = 4 \\ 2x_1 + x_2 & + x_4 & = 18 \\ & x_2 & + x_5 = 10 \end{array}$$

is in canonical form because $[A_3 \ A_4 \ A_5] = I_3$.

- Given a linear system in canonical form, we can set the identity matrix as a basis matrix. Then the corresponding basic solution can be read directly from the canonical form.

e.g.: In the above example, let

$$B = [A_3 \ A_4 \ A_5] = I_3.$$

Then x_3, x_4 and x_5 are basic variables. x_1 and x_2 are nonbasic variables.

The basic solution is: $x = (0, 0, 4, 18, 10)$.

- For better bookkeeping, we put the augmented matrices in tableaux

e.g.:

[illegible]

• Pivoting

- Q: Given a linear system in canonical form, if we slightly adjust the choice of basic variables, can we find the new canonical form easily?

A: The new canonical form can be derived by pivoting.

eg.: We consider the following canonical form again.

	x_1	x_2	x_3	x_4	x_5	RHS
x_3	1	-2	1	0	0	4
x_4	2	1	0	1	0	18
x_5	0	1	0	0	1	10

The corresponding basic variables are x_3, x_4 , and x_5 . Suppose that we want x_1 to be basic and x_4 to be nonbasic. (We also say that x_1 enters the basis and x_4 leaves the basis.) We can pivot about the element a_{21} .

	x_1	x_2	x_3	x_4	x_5	RHS
x_3	1	-2	1	0	0	4
x_4	2	1	0	1	0	18
x_5	0	1	0	0	1	10

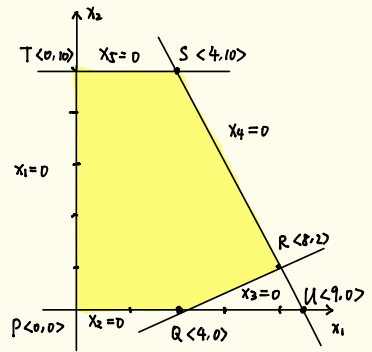
→

	x_1	x_2	x_3	x_4	x_5	RHS
x_3	0	$-\frac{5}{2}$	1	$-\frac{1}{2}$	0	-5
x_1	1	$\frac{1}{2}$	0	$\frac{1}{2}$	0	9
x_5	0	1	0	0	1	10

The current basis is $\bar{B} = [A_3, A_1, A_5]$. x_1, x_3, x_5 are basic variables, and $x = (9, 0, -5, 0, 10)$ is the current basic solution.

On the graph, the pivoting moves the corresponding basic solution from $P(0,0)$ to $U(9,0)$.

	x_1	x_2	x_3	x_4	x_5	RHS		x_1	x_2	x_3	x_4	x_5	RHS	
x_3	1	-2	1	0	0	4		x_3	0	$-\frac{3}{2}$	1	$-\frac{1}{2}$	0	-5
x_4	②	1	0	1	0	18	→	x_1	1	$\frac{1}{2}$	0	$\frac{1}{2}$	0	9
x_5	0	1	0	0	1	10		x_5	0	1	0	0	1	10



- Remark: 1. Pivoting is possible if and only if the pivot element is nonzero.

2. The tableau corresponding to a basis matrix B is in fact the augmented matrix

$$[B^{-1}A \mid B^{-1}b]$$

Pivoting from B to $\bar{B}$ is equivalent to left multiplying the invertible matrix $\bar{B}^{-1}B$ to the augmented matrix.

• The Minimum Ratio Test

In the previous example, we see that if we choose both the entering variable and departing variable arbitrarily, we may lose feasibility in the process of pivoting. Therefore, we ask the following question:

Q: Given a BFS and an entering variable, how can we choose a departing variable such that the new basic solution remains feasible?

A: The minimum ratio test.

The minimum ratio test :

Let $\beta = \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_m \end{bmatrix}$ be the vector in the RHS column of the current tableau. Let $\alpha_k = \begin{bmatrix} \alpha_{1k} \\ \vdots \\ \alpha_{mk} \end{bmatrix}$ be the vector in the k th column of the current tableau. If x_k is the entering variable, we choose α_{rk} to pivot, where

$$\frac{p_r}{\alpha_{rk}} = \min_{i \in \{1, \dots, m\}} \left\{ \frac{p_i}{\alpha_{ik}} \mid \alpha_{ik} > 0 \right\}.$$

e.g.:

	x_1	x_2	x_3	x_4	x_5	RHS
x_3	1	-2	1	0	0	4
x_4	2	1	0	1	0	18
x_5	0	1	0	0	1	10

current basis: $B = [A_3 \ A_4 \ A_5]$

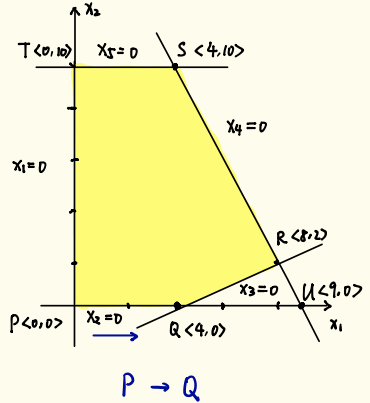
Current BFS: $\chi = (0, 0, 4, 18, 10)$

Suppose that we have decided to make x_1 basic. Then

We compare

$$\frac{\beta_1}{\alpha_{11}} = \frac{4}{1} = 4 \quad \text{and} \quad \frac{\beta_2}{\alpha_{21}} = \frac{18}{2} = 9.$$

Therefore, x_3 will be the departing variable.



	x_1	x_2	x_3	x_4	x_5	RHS
x_3	1	-2	1	0	0	4
x_4	2	1	0	1	0	18
x_5	0	1	0	0	1	10



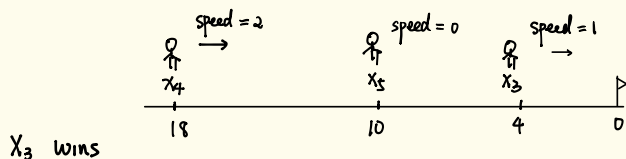
	x_1	x_2	x_3	x_4	x_5	RHS
x_1	1	-2	1	0	0	4
x_4	0	5	-2	1	0	10
x_5	0	1	0	0	1	10

new basis: $B = [A_1, A_4, A_5] = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

new BFS: $x = (4, 0, 0, 10, 10)$

Remark: 1. Intuitively, the minimum ratio test is like a racing game.

β_i is the distance between Player i and the destination. α_{ik} is the speed of Player i .

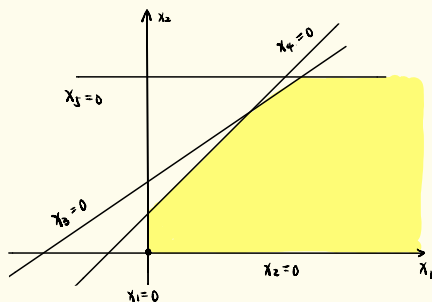


2. The minimum ratio $\min_{i \in \{1, \dots, m\}} \left\{ \frac{\beta_i}{\alpha_{ik}} : \alpha_{ik} > 0 \right\}$ is well-defined if and only if there is an index i such that

$\alpha_{ik} > 0$. If no such i exists, then $\alpha_k = \begin{bmatrix} \alpha_{1k} \\ \alpha_{2k} \\ \vdots \\ \alpha_{mk} \end{bmatrix} \leq 0$ and the feasible region is unbounded.

eg.: Consider a system with the following canonical form.

	x_1	x_2	x_3	x_4	x_5	RHS
x_3	-2	3	1	0	0	6
x_4	-1	1	0	1	0	1
x_5	0	1	0	0	1	5



$$\begin{cases} -2x_1 + 3x_2 \leq 6 \\ -x_1 + x_2 \leq 1 \\ x_2 \leq 5 \\ x_1, x_2 \geq 0 \end{cases}$$

3. If there is a tie in the minimum ratio test, we can break the tie arbitrarily. The new BFS will be a degenerate BFS.

eg.: Consider a system with the following canonical form.

	x_1	x_2	x_3	x_4	x_5	x_6	RHS
x_5	1	1	0	4	1	0	10
x_3	0	1	1	2	0	0	6
x_6	-2	2	0	3	0	1	12

Suppose that x_2 is the entering variable, then the minimum ratio test shows:

$$\frac{\theta_1}{\alpha_{12}} = \frac{10}{1} = 10, \quad \frac{\theta_2}{\alpha_{32}} = \frac{6}{1} = 6, \quad \frac{\theta_3}{\alpha_{62}} = \frac{12}{2} = 6.$$

We can choose either x_3 or x_6 as the departing variable. The resulting canonical forms are

	x_1	x_2	x_3	x_4	x_5	x_6	RHS
x_5	1	1	0	4	1	0	10
x_3	0	1	1	2	0	0	6
x_6	-2	2	0	3	0	1	12

→

	x_1	x_2	x_3	x_4	x_5	x_6	RHS
x_5	1	0	-1	2	1	0	4
x_3	0	1	1	2	0	0	6
x_6	-2	0	-2	-1	0	1	0

← degeneracy

and

	x_1	x_2	x_3	x_4	x_5	x_6	RHS
x_5	1	1	0	4	1	0	10
x_3	0	1	1	2	0	0	6
x_6	-2	2	0	3	0	1	12

→

	x_1	x_2	x_3	x_4	x_5	x_6	RHS
x_5	2	0	0	$\frac{5}{2}$	1	$-\frac{1}{2}$	4
x_3	1	0	1	$\frac{1}{2}$	0	$-\frac{1}{2}$	0
x_2	-1	1	0	$\frac{3}{2}$	0	$\frac{1}{2}$	6

← degeneracy.

respectively.

Note that they result in the same BFS (despite that they correspond to different bases.)